

Sumit Kumar

Software Development Engineer | Backend, Messaging Systems & AI Agents

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SUMMARY

SDE building AI-enabled, event-driven CPaaS systems across WhatsApp, RCS, and SMS. Strong in TypeScript/Node.js, Redis Streams, MongoDB, async processing, API gateways, webhook reliability, observability, and production backend delivery for high-volume messaging platforms.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, SQL; Go reading/debugging/testing with AI-assisted workflows

Backend: REST APIs, microservices, event-driven architecture, API gateways, webhook systems, JWT auth, rate limiting, circuit breakers, idempotency, DLQs

Data and Messaging: Redis Streams, BullMQ, Redis Pub/Sub, MongoDB, PostgreSQL, Supabase, GCP Pub/Sub

AI and Agents: Gemini API, intent classification, agent routing, prompt/context assembly, knowledge-base retrieval, response caching, AI-assisted development workflows

Infra and Observability: Docker, Nginx, Datadog APM, OpenTelemetry, Prometheus, Zap, Winston, AWS EC2/S3/CloudFront, Git/GitHub

Frameworks and Testing: Express.js, Node.js, Next.js, Go test, race tests, benchmarks, Testcontainers, miniredis, testify, Jest

EXPERIENCE

Software Development Engineer, Celetel

Noida, India | Nov 2024 - Present

- Built backend services for a multi-channel CPaaS platform across WhatsApp, RCS, SMS, webhooks, AI-agent flows, and provider delivery paths on a 2-engineer team; helped the system withstand 10M+ messages/day.
- Worked on an AI-agent flow processor for WhatsApp conversations, covering event-driven message normalization, intent classification, agent routing, business-context assembly, knowledge-base lookup, Gemini response generation, Redis-backed caching, and analytics.
- Co-delivered a high-frequency Unified Messaging Service gateway in Go with API-key auth, MongoDB-backed key validation, Redis auth caching, request tracing, circuit breakers, health probes, and async 202 Accepted publishing to Redis Streams.
- Implemented single and bulk messaging APIs supporting up to 1,000 messages/request; documented gateway benchmarks of roughly 32.5K single-message req/s and 350K bulk msgs/s on single-core Redis-backed tests.
- Co-delivered a Go UMS processor using Redis Streams with XREADGROUP, priority draining, XACK, pending-entry reclaim, DLQ handling, transformation, provider resolution, template validation, opt-out filtering, batching, and outbound stream publishing.
- Added idempotent MongoDB BulkWrite persistence before delivery fanout, preventing orphaned delivery events; measured UMS processor throughput at roughly 31.5K full round-trip msgs/s single-threaded and about 94K msgs/s with three consumer goroutines.
- Migrated webhook processing from Supabase-backed retry flows toward GCP Pub/Sub, improving burst reliability while reducing infrastructure cost and operational load; preserved rollback paths through an additive raw-webhook consumer for Meta WhatsApp, Sinch RCS, and SMS callbacks.
- Designed disaster-recovery handling for frequent data-center network fluctuations and added production operability: idempotency, deduplication windows, structured logs, Prometheus/Datadog metrics, trace propagation, readiness/liveness probes, graceful shutdown, race tests, benchmarks, Testcontainers, and miniredis tests.
- Integrated with coworker-owned notification and IAM services through service contracts, webhook/event flows, auth boundaries, and operational handoffs.

Product Management Intern, STAGE

Gurugram / Remote | Mar 2024 - Sep 2024

- Proposed and implemented retention experiments; supported product analysis, user-feedback aggregation, market research, and cross-functional planning with engineering and business stakeholders.

SELECTED PROJECTS

cGPT - Document-Aware Company Chatbot

GitHub: github.com/thisumitk/cGPT

- Built an Express.js chatbot backend using OpenAI, LangChain, FAISS vector search, MongoDB, UUID-based sessions, and document-context retrieval.

EDUCATION

Bachelor of Business Administration (Hons.), ITM University Gwalior | 2020 - 2023

Self-directed CS and backend engineering: CS50, The Odin Project, FreeCodeCamp, production codebase work, distributed systems and backend architecture practice.